

Mini Lead Management System

Overview

Mini Lead Management System is a role-based lead management application built using:

- Node.js
- Express.js
- PostgreSQL
- JWT Authentication

The system allows Admins, Managers, and Agents to manage leads efficiently with secure authentication and activity tracking.

Tech Stack

Backend

- Node.js
- Express.js
- PostgreSQL
- JWT Authentication
- bcrypt
- dotenv

Database

- PostgreSQL
-

Architecture

The project follows a layered architecture:

```
src
|
├─ routes
├─ controllers
├─ services
├─ repositories
├─ middleware
├─ utils
└─ config
```

Layers

Routes

Responsible for API endpoint definitions.

Controllers

Handle incoming requests and responses.

Services

Contain business logic.

Repositories

Handle database interactions.

Middleware

Authentication and authorization.

Authentication Module

Features

- User Registration
- User Login
- JWT Authentication
- Protected Routes
- Role-Based Access Control

Roles

Admin

- Create Leads
- Update Leads
- Delete Leads
- Assign Leads
- View All Leads

Manager

- Create Leads
- Update Leads
- Assign Leads
- View All Leads

Agent

- View Assigned Leads
 - Update Lead Status
-

Database Schema

Users

Field	Type
id	SERIAL
full_name	VARCHAR
email	VARCHAR
password_hash	TEXT
role	VARCHAR
is_active	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP

Leads

Field	Type
id	SERIAL
name	VARCHAR
email	VARCHAR
phone	VARCHAR
source	VARCHAR
status	VARCHAR
notes	TEXT
assigned_to	INTEGER
created_by	INTEGER
created_at	TIMESTAMP
updated_at	TIMESTAMP

Activity Logs

Field	Type
id	SERIAL
lead_id	INTEGER
user_id	INTEGER
activity_type	VARCHAR
old_value	JSONB
new_value	JSONB
created_at	TIMESTAMP

Authentication APIs

Register User

`POST /api/auth/register`

Request

```
{
  "fullName": "John Doe",
  "email": "john@example.com",
  "password": "123456",
  "role": "AGENT"
}
```

Login User

`POST /api/auth/login`

Request

```
{
  "email": "john@example.com",
  "password": "123456"
}
```

Response

```
{
  "accessToken": "jwt-token"
}
```

Get Profile

```
GET /api/auth/profile
```

Authorization:

```
Bearer <token>
```

Lead APIs

Create Lead

```
POST /api/leads
```

Get All Leads

Supports:

- Pagination
- Search
- Sorting
- Filtering

Example:

```
GET /api/leads?page=1&limit=10
```

```
GET /api/leads?search=john
```

```
GET /api/leads?status=NEW
```

```
GET /api/leads?source=Website
```

```
GET /api/leads?sortBy=created_at&sortOrder=desc
```

Get Lead By Id

```
GET /api/leads/:id
```

Update Lead

```
PUT /api/leads/:id
```

Update Lead Status

```
PATCH /api/leads/:id/status
```

Example:

```
{
  "status": "CONTACTED"
}
```

Assign Lead

```
PATCH /api/leads/:id/assign
```

Example:

```
{
  "assignedTo": 3
}
```

Delete Lead

```
DELETE /api/leads/:id
```

Activity Logs

Get All Activities

```
GET /api/activity
```

Get Activity By Lead

```
GET /api/activity/:leadId
```

Auto Assignment Logic

When a Manager creates a lead:

1. System finds active agents.
2. Lead is automatically assigned.
3. Activity log is created.

Current implementation supports automatic lead assignment.

Activity Tracking

The system tracks:

- LEAD_CREATED
- LEAD_UPDATED
- LEAD_ASSIGNED
- STATUS_CHANGED

All activities are stored in the `activity_logs` table.

Environment Variables

```
DB_HOST=localhost
DB_USER=root
DB_PASSWORD=root
DB_NAME=lead_management_system
DB_PORT=5432

PORT=3000

JWT_SECRET=your-secret-key

ACCESS_TOKEN_EXPIRY=15m
```

Running the Project

Install dependencies:

```
npm install
```

Start server:

```
npm run dev
```

Server:

```
http://localhost:3000
```

Implemented Requirements

Authentication

- Register User
- Login User
- JWT Authentication
- Password Hashing
- Protected APIs
- Role-Based Authorization

Lead Management

- Create Lead
- Update Lead
- Delete Lead
- Get Lead By ID
- List Leads

Lead Listing Features

- Pagination
- Search
- Sorting
- Status Filtering
- Source Filtering

Activity Logs

- Lead Created
- Lead Updated
- Lead Assigned
- Status Changed

Future Enhancements

- Refresh Token Support
- Email Notifications
- Third Party Integrations
- Dashboard Analytics
- Frontend Application (React.js)

Developed as part of the Mini Lead Management System Assignment.